AHAMMAD SHAWKI

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) Expected 2028
BSc in Computer Science and Engineering *Dhaka, Bangladesh*
Coursework: Data Structures and Algorithms I and II, Database, Software Engineering, Computer Architecture, Theory of Computation.

EXPERIENCE

Founder and Developer, SrotDev May 2025 to Present

- Lead an engineering collective shipping production web and mobile applications across React, Next.js, Django, FastAPI and PostgreSQL.
- Architected CareForAll as seven independently deployable Django services behind an API gateway, with its own Prometheus and Grafana observability stack.
- Built Srabon, a gamified science learning platform with multilingual and accessibility support, as the team's flagship product.

President of Web and App Development, NDITC Mar 2022 to Jun 2024

- Led a 30-person engineering team; delivered the rebuilt official website and the UI/UX for the club's native mobile application.
- Raised over 1,000,000 BDT in sponsorship across 100+ company negotiations for INIT, the club's national annual event.
- Taught web development fundamentals directly to more than 500 students as a workshop instructor.

Founder and Team Lead, The AS8 Organization Feb 2021 to May 2022

- Built a developer community that reached more than 20,000 learners through free programmes, courses and technical content.
- Partnered with Codecademy, Hack Club and DeepLearning.AI; wrote articles published on freeCodeCamp News.

PROJECTS

DeepSIFT, auditable digital forensics MCP, Python, Volatility 3, MITRE ATT&CK

- 148-tool Model Context Protocol layer that parses forensic binaries into structured JSON before the model reads them, with a signable chain of custody. Scored 4/4 against published ground truth on two cases, zero hallucinations, 100% claim grounding.

TRACE, camera heart rate with no trained model Python, NumPy, classical signal processing

- Recovers a pulse from the sub-1% colour change a heartbeat makes in webcam video using only convolution and the Fourier transform, switching between three published methods every half second on spectral confidence. 4.62 BPM mean absolute error over 20 volunteers against 5.63 for the best single method.

Cascade, agent memory that expires on its own Python, AWS, vector search

- Compiles a resolved incident into a runbook pinned to the version of every policy rule it consulted, so editing one rule invalidates everything derived from it in four writes. Reuse calls no model: retrieval is an index, freshness a join.

MACHINE LEARNING COMPETITIONS

CADENCE, voice-based Parkinson's screening ML Empowerment Build Challenge, 2026

- Won best overall project. Screens from a 30-second voice sample and explains every score, validated across three independent language corpora using cross-database testing and domain-adversarial training to avoid microphone bias.

Olikobochon, Bengali LLM hallucination detection IUTCS Datathon 2.0, 2026

- Third on the public leaderboard at 0.922 F1 on the hallucinated class. A decision ladder resolves most rows deterministically against source corpora and an 89k-question bank, sending only the residual to an open-weight judge with Wikipedia retrieval.

DaT Parkinson's Challenge, SPECT scan classification DrivenData, 2026

- Tenth on the private leaderboard at 0.2760 log loss, second highest AUROC in the top ten at 0.9551. Registers the striatum to a template built from normal scans with no atlas, then stacks regional binding ratios with 2D and slice-sequence networks.

AWARDS

Grand Prize Winner, FLIR App Challenge 2025 to 2026, for SolarSnap. **Winner, Golden Agent**, Everyday Agents category, Agents for Humans Hackathon, 2026, for Shoulder. **Best Overall Project**, ML Empowerment Build Challenge 2.0, 2026, for CADENCE. **Best ERNIE Multimodal Application**, ERNIE AI Developer Challenge by Baidu, 2026, for Doclyst. **Honourable Mention**, LMA EDGE Hackathon, 2026, presented at the London finale, for Coven.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, Dart, HTML/CSS

Frameworks: React, Next.js, Django, Django REST Framework, FastAPI, Flask, Node.js, Flutter, JavaFX

AI and Machine Learning: PyTorch, scikit-learn, Model Context Protocol, Google ADK, Retrieval Augmented Generation, Whisper, Gemini, Claude, Pandas, NumPy

Infrastructure: Docker, Google Cloud (Cloud Run, Pub/Sub, Firestore, Vertex AI), AWS (Bedrock, ECS, Lambda), PostgreSQL, MongoDB, Redis, GitHub Actions, Grafana